MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL
(Formerly West Bengal University of Technology)
Syllabus of BCA
(Effective from 2023-24 Academic Sessions)

SEMESTER: IV

DEFINITION OF CREDIT

1 HR LECTURE PER WEEK	1 CREDIT
1 HR TUTORIAL PER WEEK	1CREDIT
2 HR PRACTICAL PER WEEK	1 CREDIT

SUBJECT NUMBERING SCHEME:

CODE FOR THE DEPT. OFFERING SUBJECT	SUBJECT TYPE	SEM	SUBJECT CODE

C	CORE MAJOR

SUBJECT NAME: Data Base Management System **Credit: 5 (3L + 2P)**
SUBJECT CODE: BCAC401

COURSE OBJECTIVE:

The course on Database Management Systems (DBMS) aims to equip students with a comprehensive understanding of the core principles and practical applications of managing data effectively within organizational contexts. Throughout the course, students will delve into the fundamental concepts of database design, data manipulation, and database administration. They will learn to design efficient and robust databases using entity-relationship modeling and normalization techniques, ensuring data integrity and optimal performance. By the end of the course, students will be proficient in designing, implementing, and managing databases to support decision-making processes and organizational objectives effectively.

COURSE OUTCOME	
CO1	Understand the fundamental concepts and principles of database management systems (DBMS), including data modeling, database design, and normalization.
CO2	Demonstrate proficiency in using SQL (Structured Query Language) to perform data definition, manipulation, and query tasks in a relational database environment.

CO3	Apply indexing and query optimization techniques to enhance the performance of database systems
CO4	Evaluate different types of database architectures, such as centralized, distributed, and client-server architectures, and comprehend their advantages and limitations.

DETAILED SYLLABUS:

Module No:	NAME OF THE TOPIC	HOURS	L	T	MARKS
M1	Introduction: Concept & Overview of DBMS, Data Models, Database Languages, Database Administrator, Database Users, Data Abstraction, Three Schema architecture of DBMS.	2	2		2
M2	Entity Relationship Model: Entity Set, Simple and composite Attribute, Single valued and multivalued attribute, Relatonship sets, Mapping cardinality, keys, Binary vs n-ary relationship, Entity Relationship Diagram : Need for E-R Model, Various steps of database design, Mapping Constraints, E-R diagram, Subclass, Generalization, Specialization, Aggregation, Strong Entity-Weak Entity, Tabular representation of Strong entity set, tabular representation of weak entity set, ER diagram and corresponding UML diagram	5	4	1	10
M3	Relational Algebra: Select operation, Project Operation, Set operations (union, intersection, difference), Join operations, Division operation, outer join and outer union, Examples queries in Realtional Algebra	4	3	1	8
M4	SQL : Concept of DDL, DML. Basic Structure Relational databases and tables, Set operations, Aggregate Functions, Null Values, Domain Constraints, Referential Integrity Constraints, assertions, views, Nested Subqueries, Stored procedures,cursors and triggers.	4	3	1	10
M5	Relational Model and Relational Database Design: Concept of Relational Model, Design Issues, Keys, Closure set, Functional Dependency, Different anomalies in designing a Database., Normalization using functional dependencies, Decomposition, Boyce-Codd	10	7	3	14

	Normal Form, 3NF, Normalization using multivalued dependencies, 4NF,5NF.				
M6	Indexing and Hashing: Ordered indices (Primary Index, Dense and Sparse Indices), Secondary Index, B tree and B+ tree indexing, Hashing Concepts and its implementation Basic query optimization techniques, Cost estimation in query optimization	10	7	3	14
M7	Transaction Management: Transaction definition, properties, transaction state diagram, commit and rollback, Serializability (Conflict and View), Concurrency control, lock based protocols, Two phase locking, Timestamp ordering protocol, Recovery management, Deadlock handling and prevention	10	8	2	12
	INTERNAL EXAMINATION	3			30
	TOTAL	48	34	11	100

Practical:

SUBJECT NAME: DBMS Lab **Credit: 2**
SUBJECT CODE: BCAC491

Use any database for the SQL implementation

List of sample Questions for Practical:

1. Write a SQL query to retrieve all columns from a table named "Employees."
2. How do you select distinct values from a column in SQL?
3. Write a SQL query to retrieve all records from the "Orders" table where the order amount is greater than 1000.
4. Explain the difference between INNER JOIN and LEFT JOIN in SQL with an example.
5. How do you use the WHERE clause to filter rows in SQL?
6. Write a SQL query to calculate the total number of orders for each customer from the "Orders" table.
7. What is the purpose of the GROUP BY clause in SQL? Provide an example query.
8. How do you use the HAVING clause in SQL? Provide an example.
9. Write a SQL query to update the salary of an employee with ID 101 to 50000.
10. How do you delete records from a table in SQL? Provide an example.
11. Explain the concept of foreign keys in SQL with an example.
12. Write a SQL query to retrieve the top 5 highest-paid employees from the "Employees" table.
13. How do you use the ORDER BY clause in SQL? Provide an example query.
14. Explain the difference between the UNION and UNION ALL operators in SQL.

15. Write a SQL query to find the average salary of all employees.
16. How do you use the LIKE operator in SQL? Provide an example.
17. Write a SQL query to retrieve the names of all employees whose names start with the letter 'A.'
18. Explain the concept of indexes in SQL databases.
19. Write a SQL query to find the maximum and minimum salary from the "Employees" table.
20. How do you use the BETWEEN operator in SQL? Provide an example.

Based on the curriculum as covered by the subject teacher

SUGGESTED READING:

1. "Database Management Systems" by Raghu Ramakrishnan, Johannes Gehrke - Publisher: McGraw-Hill Education
2. "Database System Concepts" by Abraham Silberschatz, Henry F. Korth, S. Sudarshan - Publisher: McGraw-Hill Education
3. "Fundamentals of Database Systems" by Ramez Elmasri, Shamkant B. Navathe - Publisher: Pearson
4. "Database Management Systems: Designing and Building Business Applications" by Gerald V. Post - Publisher: Wiley
5. "Database Systems: The Complete Book" by Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom - Publisher: Pearson
6. "Principles of Database Management" by Wilfred Hansen - Publisher: Cengage Learning
7. "Database Management Systems" by Ivan Bayross - Publisher: BPB Publications

SUBJECT NAME: Operating System **Credit: 4 L (3L +1 T)**
SUBJECT CODE: BCAC402

COURSE OBJECTIVE:

The course on Operating Systems is designed to provide students with a comprehensive understanding of the fundamental principles and functionalities underlying modern computer operating systems. Throughout the course, students will explore key concepts such as process management, memory management, file systems, and I/O management. They will learn about the role of the operating system in resource allocation, scheduling, and synchronization, gaining insights into how these mechanisms contribute to efficient and reliable system operation.

COURSE OUTCOME	
CO1	Understand the fundamental concepts and functionalities of operating systems, including process management, memory management, file systems, and I/O management.
CO2	Analyze and compare different types of operating systems, such as batch processing systems, time-sharing systems, real-time systems, and distributed systems, in terms of their design principles, advantages, and limitations.
CO3	Evaluate different I/O management strategies, including buffering, caching, and device drivers, to optimize the performance and reliability of I/O operations in the operating system.
CO4	Apply memory management techniques, such as paging, segmentation, and virtual memory, to efficiently utilize the system's memory resources and provide a logical abstraction of memory to processes.
CO5	Collaborate effectively in teams to analyze, design, and implement operating system components and solutions for real-world scenarios, demonstrating effective communication and problem-solving skills.

DETAILED SYLLABUS:

Module No:	NAME OF THE TOPIC	HOURS	L	T	MARKS
M1	Computer H/w review (Processors, Memory, devices, I/O bus), Operating system Basic concepts, Architecture of OS, Introduction of Different types of Operating System(Mainframe, server side OS Multiprocessor OS, Embedded OS, Real Time OS, Sensor node OS, Smart card OS), Virtualization, Free and Open Source Operating System	3	2	1	5
M2	Operating System Architecture (Monolithic System, Layered System, Microkernel, client Server model, System Calls, Linker and Loader, Booting of an Operating System	2	1	1	5
M3	Process Management: Process, Process State Diagram, Process Control Block, Process Scheduling criteria, Process scheduling algorithms, Types of schedulers, threads, types of thread, Thread Scheduling, Inter Process Communication , Race Condition, Critical region, use of Semaphore, mutex, and monitor, Classical problems on Synchronization	18	14	4	18

M4	Deadlock: Deadlock Characterization, Methods of handling Deadlock, Deadlock prevention and avoidance, deadlock detection and Recovery from deadlock	4	3	1	8
M5	Memory Management: Continuous Memory Allocation, Paging, Swapping, Virtual memory: Paging, Page table Structure, Page Table for large memory, Page replacement Algorithms, page Size, Page Fault Handling, Segmentation	14	10	4	15
M6	File management: File naming, File structures, File Types, Single Level and Hierarchical OS, Shared file, Disk managent and related algorithms	4	3	1	8
M7	Security and Protection: security Threats and Attackers, Controlling Access to Resources (Protection Domain, access Control List,) Exploiting Software (Buffer overflow attack, Integer Overflow attack, Dangling Pointer, Null Pointer Dereference Attack), Malware (Worm, virus and Trojan)	8	6	2	6
M8	Distributed Operating System: Goal of Distributed OS, Remoter Procedure call, Name resolution, Clock Synchronization	4	3	1	5
	INTERNAL EXAMINATION	3			30
	TOTAL	60	42	15	100

SUGGESTED READING:

1. "Operating System Concepts" by Abraham Silberschatz, Peter Baer Galvin, Greg Gagne - Publisher: Wiley
2. "Modern Operating Systems" by Andrew S. Tanenbaum, Herbert Bos - Publisher: Pearson
3. "Operating Systems: Internals and Design Principles" by William Stallings - Publisher: Pearson
4. "Operating System Design and Implementation" by Andrew S. Tanenbaum, Albert S. Woodhull - Publisher: Pearson
5. "Operating Systems: Principles and Practice" by Thomas Anderson, Michael Dahlin - Publisher: Recursive Books
6. "Operating Systems: Three Easy Pieces" by Remzi H. Arpaci-Dusseau, Andrea C. Arpaci-Dusseau - Publisher: Arpaci-Dusseau Books
7. "Modern Operating Systems: Global Edition" by Andrew S. Tanenbaum, Herbert Bos - Publisher: Pearson

SUBJECT NAME: Software Engineering **Credit: 4 (3L + 1T)**
SUBJECT CODE: BCAC403

COURSE OBJECTIVE:
The course on Software Engineering aims to provide students with a comprehensive understanding of the principles, methodologies, and best practices essential for developing high-quality software systems. Throughout the course, students will delve into various aspects of the software development lifecycle, including requirements analysis, design, implementation, testing, deployment, and maintenance. By the end of the course, students will be proficient in designing, developing, and managing software systems that meet stakeholder requirements and industry standards.

COURSE OUTCOME	
CO1	Understand the fundamental principles and concepts of software engineering, including software development life cycle models, requirements engineering, and software design paradigms.
CO2	Analyze and apply various software development methodologies, such as Waterfall, Agile, Scrum, and DevOps, to plan, execute, and manage software projects effectively.
CO3	Apply software design principles, such as modularity, abstraction, encapsulation, and cohesion, to create maintainable, scalable, and extensible software solutions.
CO4	Perform software testing activities, including unit testing, integration testing, system testing, and acceptance testing, to ensure the quality, reliability, and correctness of software products.
CO5	Collaborate effectively in multidisciplinary teams to analyze, design, implement, and test software solutions for real-world problems, demonstrating effective communication, teamwork, and problem-solving skills

DETAILED SYLLABUS:

Module No:	NAME OF THE TOPIC	HOURS	L	T	MARKS
M1	Introduction: A Generic View of Software Engineering, Phases in software development, Linear Sequential Model, Prototype model, Evolutionary Model (Incremental and spiral model), Specialized Process Model (Component-Based Development, Aspect-Oriented Software Development), Agile Process (Principles, Human factors), Simple Case Study	4	3	1	6

M2	Project Metrics: Software Measurement (Size oriented, Function Oriented, Extended Function Point Metrics, Object-Oriented Metrics, Web application project Metric), metric for Software quality (Measuring Quality, Defect Removal efficiency), Integrate metric with software(Establishing a baseline)	4	3	1	5
M3	Object Oriented Concepts: class, Objects, attributes, Operations, Methods, and Services, Messages, Identifying the Elements of an Object Model (identification of class and objects, Defining Operations), Object Oriented analysis and Design (use cases, Class-Responsibility-Collaborator Modeling,), Object-Relationship Model	4	3	1	6
M4	Project planning : Identification of Software scope (Feasibility), Resource Identification (Human resource, Reusable Software Resources), Empirical Estimation Models (COCOMO model), Estimation for Object oriented project, Estimation for agile development, Estimation for Web application project, Estimation of human resource requirement, Team structure, Time estimation, Project scheduling (Time-Line Charts, Tracking the Schedule, Scheduling for WebApp Projects) Make / buy decision (Creating a Decision Tree), Project Monitoring Plan (Time sheet, reviews, Cost schedule Milestone graph), Risk management (Identification, Prioritization, Risk Mitigation, Monitoring, and Management)	15	12	3	15
M5	Software Architecture (Fan out and fan in structure), Structural Partitioning (Horizontal and vertical Partition), Functional Independence (Coupling and Cohesion), Design heuristics for effective modularity	2	2	0	4
M6	Design: data Design , Architectural design / mapping using data Flow(Transform flow and Transactional flow), Designing class Based Components, Component Level design for Web Application, User Interface design Technique and documentation (Reduce the User's Memory Load, make interface consistent), Interface Design steps (User Interface Design Patterns, design issues), Discuss with a case study,	8	6	2	6

M7	Software Testing : Objective, Pimples, Test case Design for conventional software (Unit testing, Integration Testing), path testing , cyclometic complexity, Test Strategy for Object Oriented software, Test cases for web application, validation testing, System testing (Recovery testing, Security testing, stress testing, Performance testing)	10	7	3	14
M8	Quality Management: Define quality of software, McCall's Quality Factors, ISO 9126 Quality Factors, Achieve software quality (Software Engineering Methods, project Management Technique), Quality Control, Quality Assurance (elements of quality assurance), SQA goals, tasks, Metrices, Six Sigma for Software Engineering	6	5	1	6
M9	Software Configuration Management (SCM): Elements of a Configuration Management System, Baseline, Software Configuration Items, SCM Features, SM processes (Version Control, Change Control, Configuration Audit, Status Reporting), SCM for Web application (WebApp Configuration Objects, Content management, Change management)	4	3	1	8
	INTERNAL EXAMINATION	3			30
	TOTAL	60	44	13	100

SUGGESTED READING:

1. "Software Engineering: Principles and Practices" by Deepak Jain, S. K. Gupta - Publisher: Laxmi Publications
2. "Software Engineering and Quality Assurance" by Kshirasagar Naik, Priyadarshi Tripathy - Publisher: Oxford University Press
3. "Software Engineering: Theory and Practice" by Shariq Mahmood, A. A. Sastry - Publisher: Oxford University Press
4. "Software Engineering: A Precise Approach" by Pankaj Jalote - Publisher: Wiley India
5. "Software Engineering" by Pankaj Jalote - Publisher: Pearson Education India
6. "Software Engineering: A Lifecycle Approach" by Surajit Ghosh, Anirban Basu - Publisher: Pearson Education India
7. "Fundamentals of Software Engineering" by Rajib Mall - Publisher: Prentice Hall India
8. "Software Engineering: A Practitioner's Approach" by Roger S. Pressman - Publisher: McGraw-Hill Education
9. "Software Engineering" by Ian Sommerville - Publisher: Pearson Education Limited
10. "Introduction to the Team Software Process" by Watts S. Humphrey - Publisher: Addison-Wesley Professional

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Course: Society Culture and Human Behavior	
Course Code: AECC401A	**Semester: IV**
	Maximum Marks: 100
Teaching Scheme	**Examination Scheme**
Lecture: 2	End semester Exam: 70
Tutorial: 0	Attendance: 5
Practical: 0	Continuous Assessment: 25
Credit: 2	Practical/Sessional internal continuous evaluation: 0
	Practical/ Sessional external examination: 0

Sl. No.	**Course Objective**	
1	To explore the relationship between society, culture and human behaviour	
2	To analyse the impact of social norms, values and beliefs on individual and collective behaviour	
3	To examine the cultural diversity and its influence on social interactions and perceptions	
	Course Outcomes	**Mapped module/Unit**
CO 1	Demonstrate and understanding of the interplay between society, culture and human behaviour	U1, U2
CO 2	To know about caste system, unemployment and poverty	U3, U4
CO 3	Critically assess theories and concept related to human behaviour	U5

Learning Outcome/Skills:

The candidate is able to have a detailed understanding of the importance of society, culture and human behavior which are high required to live in a society.

Unit	**Total Hours**	**% of Questions**	**Bloom's Taxonomy**	**Remarks, if any**
THEORY				
U1	**6**	**20**	**1, 2, 3**	**NA**

U2	**6**	**20**	**1, 2, 3**	**NA**
U3	**6**	**20**	**1, 2, 3**	**NA**
U4	**6**	**20**	**1, 2, 3**	**NA**
U5	**6**	**20**	**1, 2, 3**	
	30	**100%**		

Course Code:	**AECC401A**	
Course:	**Society Culture and Human Behavior** Credits:2.0	
Contents		
Chapter	**Name of the topic**	**Hours**
Unit-I	Demographic Profile: Characteristics of Indian Population, Population Growth, Age, Sex, Religion, Language, Occupations, National Policy on Population	**6**
Unit-II	Indian Society and culture: Society and its types, Culture – Features, Characteristics and Diversity. Differences with Western Culture,	**6**
Unit-III	Social Stratification: Caste System, Class System, Communities, Ethnic Groups, Weaker Section and Minorities, Constitutional Provisions for Scheduled Castes, Scheduled Tribes and other Backward Classes.	**6**
Unit-IV	Socio-Economic Problems: Poverty, Illiteracy, Unemployment, Housing, Child Labour, Migration, Occupational Diseases, Insurgency, Terrorism, Crime, Project Affected People, Social Destitute, Beggary, Aged Population, Juvenile Delinquency, Problems in Family Life.	**6**
Unit-V	**Introduction to Human Behaviour:** Overview of human behaviour, Importance of studying human behaviour, determinants of human behaviour	**6**
	Total	**30**

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List of Books

Name of Author	Title of the Book	Name of the Publisher
Andre Beteille	Society and Politics in India	OUP
Dipankar Gupta	Social Stratification	OUP
Ram Ahuja	Social Problems in India	Rawat Publications
M.N. Srinivas	Social Structure and Caste and Other Essays	OUP
A.N. Tripathi	Human Values	New Age International
NCERT	Text Book on Indian Society	NCERT
R. Thapar (ed.)	Tribe, Caste and Religion in India	Macmillian

Maulana Abul Kalam Azad University of Technology, WB
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AE Course: Universal Human Values

Code: AECC 401B

Credits: 2

Duration: 30 hours

Course Objectives: The course is designed to develop a holistic perspective based on self-exploration about themselves (human being), family, society and nature/existence and to understand (or developing clarity) the harmony in the human being, family, society and nature/existence. It also aims to strengthen self-reflection, commitment and courage to act.

Course Outcomes:

1. Demonstrate self- awaress, and awareness about their surroundings (family, society, nature)
1. Define life responsibilities in handling problems with sustainable solutions, while keeping human relationships and human nature in mind
2. Show critical ability
3. Outline commitments towards human values, human relationship and society
4. Make use of human values in different day-to-day settings in real life

Unit 1 : INTRODUCTION -

NEED, BASIC GUIDELINES FOR VALUE EDUCATION (6 hours)

Purpose and motivation for the course, recapitulation from Universal Human Values-I

Self-Exploration–what is it? - Its content and process; 'Natural Acceptance' and Experiential s

Right understanding, Relationship and Physical Facility- the basic requirements for fulfilment of aspirations of every human being with their correct priority

Understanding Happiness and Prosperity correctly- A critical appraisal of the current scenario

living in harmony at various levels.

Practice sessions to discuss natural acceptance in human being as the innate acceptance for living with responsibility (living in relationship, harmony and co-existence) rather than as arbitrariness in choice based on liking-disliking.

Unit 2 : UNDERSTANDING HARMONY IN THE HUMAN BEING - HARMONY IN MYSELF!
(6 Hours)

Understanding human being as a co-existence of the sentient 'I' and the material 'Body'

Understanding the needs of Self ('I') and 'Body' - happiness and physical facility

Understanding the Body as an instrument of 'I' (I being the doer, seer and enjoyer)

Understanding the characteristics and activities of 'I' and harmony in 'I'

Understanding the harmony of I with the Body: Sanyam and Health; correct appraisal of Physical needs, meaning of Prosperity in detail

Programs to ensure Sanyam and Health.

Practice sessions to discuss the role others have played in making material goods available to oneself. Identifying from one's own life. Differentiate between prosperity and accumulation. Discuss programs for ensuring health vs dealing with disease.

Unit 3 UNDERSTANDING HARMONY IN THE FAMILY AND SOCIETY- HARMONY IN HUMAN-HUMAN RELATIONSHIP (6 Hours)

Understanding values in human-human relationship; meaning of Justice (nine universal values in relationships) and program for its fulfillment to ensure mutual happiness; Trust and Respect as the foundational values of relationship

Understanding the meaning of Trust; Difference between intention and competence

Understanding the meaning of Respect, Difference between respect and differentiation; the other salient values in relationship

Understanding the harmony in the society (society being an extension of family): Resolution, Prosperity, fearlessness (trust) and co-existence as comprehensive Human Goals

Undivided Society, Universal Order- from family to world family.

Practice sessions to reflect on relationships in family, hostel and institute as extended family, real life examples, teacher-student relationship, goal of education etc. Gratitude as a universal value in relationships..

Unit 4 : UNDERSTANDING HARMONY IN THE NATURE AND EXISTENCE - WHOLE EXISTENCE AS COEXISTENCE (6 Hours)

Understanding the harmony in the Nature

Interconnectedness and mutual fulfillment among the four orders of nature- recyclability and self regulation in nature

Understanding Existence as Coexistence of mutually interacting units in all-pervasive space

Holistic perception of harmony at all levels of existence.

Practice sessions to discuss human beings as cause of imbalance in nature (film "Home" can be used), pollution, depletion of resources and role of technology etc.

Unit - 5 : IMPLICATIONS OF THE ABOVE HOLISTIC UNDERSTANDING OF HARMONY ON PROFESSIONAL ETHICS (6 Hours)

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Paper code: MIM401A

Mode: Offline **Credits: 4(3L+1T)**

HUMAN RESOURCE MANAGEMENT

Aim of the Course: The objective is to attain a comprehensive understanding of management.

Course Objectives: The course is designed to foster comprehension of the core principles of management. It also covers operational aspects. Upon finishing this course, students should have a grasp of the foundational tenets of management.

Goals:

After the completion of this course the students will be able to -

CO1: Summarize the overview of human resource Management.

CO2: Relate the objectives of Human Resource Planning its objectives.

CO3: Discover the concept of HRD its different objectives etc.

CO4: Elaborate the emerging areas of International Human Resource Management.

CO5: Students will apply the theoretical approach in practical field.

Sl	Course content	Mapped modules	Hour allotted
CO1	Human Resource Management-Overview Introduction of the paper, Definition of Human Resource, Definition & Concept of Personnel Management, Comparison between Personnel Management & HR. Nature, Aim and Objectives, Scope & Coverage & Nature of HRM, Importance of Human Resource Management. Historical Perspective & Evolution of Human Resource Management in India. Development of HR Functions, Structure & Function of HR Manager, Role of Line Managers in Managing Human Resources. Difference Between Line Function and Staff Function. Changing Function of Human Resource Management with Examples.	M1	8
CO2	Human Resource Planning Meaning, Objectives, Importance of Human Resource Planning, Need for HR Planning, Assessment of Available HR in the Organization, Work Load Analysis, Manning Norms, Demand Analysis of Future Requirement of HR, HR Policy	M2	6
CO3	Job Analysis: Concept, Uses, Job Description, Job Specification, Methods of collecting Job Analysis Data, Job Evaluation.	M3	8
CO4	Talent Acquisition and Training: Recruitment: Definition, Sources of Selection, Process of Selection, Difference Between Recruitment and Selection. Training: Definition, Difference between Training, Development and Education, Different Methods of Training, Training needs assessment – KIRK-PATRICK, CIPO, CIRO, Training calendar	M4	7
CO5	HRD: Definition, objective, process of HRD, Assessment of HRD Needs, HRD Methods	M5	5

CO6	Introduction to Performance appraisal: Purpose, Methods, Appraisal instruments, 360-degree Appraisal, HR Score Card, Errors in appraisal, Potential Appraisal, Appraisal Interview. Compensation Management – Calculation of wage and salary (only theory)	M6	6

Learning Outcome/ Skills:

Now a day every organization is having complex organization structures. Student will study how different departments and functions within an organization interacts, and how the organizational structure affects decision-making and efficiency.

Module Number	**Content**	**Total Hours**	**% of questions**	**Bloom Level (applicable)**	**Remarks, if any**
THEORY					
M1	Human Resource Management-Overview	8	**20**	**1,2**	**NA**
M2	Human Resource Planning	6	**20**	**1,2,3**	**NA**
M3	Job Analysis: Concept, Uses, Job Description, Job Specification, Methods of collecting Job Analysis Data, Job Evaluation.	8	**20**	**1,2**	**NA**
M4	Talent Acquisition and Training	7	**10**	**1,2,3**	**NA**
M5	HRD: Definition,	5	**10**	**1,2,3**	**NA**
M6	Introduction to Performance appraisal	6	**20**	**1,2,3**	**NA**
Total Theory		**40**	**100**		
TUTORIAL		**8**			
TOTAL		**48**			

M7	**Channel Design and management**	4	**10**	**1,2**	**NA**
M8	**Physical Distribution & Logistics**	3	**5**	**1,2**	**NA**
Total Theory		**40**	**100**		
TUTORIAL		**8**			
TOTAL		**48**			

Course Name: E-Commerce
MIM-402B(Minor)

Mode: Offline **Credits: 4(3T+1T)**

Aim of the course: The aim of an E-Commerce course is to provide students with a comprehensive understanding of electronic commerce and its various components, applications, and implications in today's digital economy.

Course Objectives: students will be equipped with the knowledge, skills, and competencies needed to navigate the dynamic and competitive landscape of e-commerce and pursue careers in various e-commerce-related roles.

Goals:

CO1: Define electronic commerce (e-commerce) and its various forms and applications.

CO2: Analyze different e-commerce business models, such as B2C, B2B, C2C.

CO3: Explore the technological infrastructure and tools required to support e-commerce operations, including website development platforms, payment gateways, and content management systems.

CO4: Understand how to develop and implement effective marketing campaigns to attract and retain customers.

CO5: Understand the legal and regulatory frameworks governing e-commerce, including consumer protection laws, privacy regulations, and intellectual property rights.

Sl	Course content	Mapped Module	Hours allotted
CO1	Introduction to ecommerce: Meaning and concept of ecommerce, ecommerce vs e-business, advantages and disadvantages of ecommerce, value chain in ecommerce, Porter's value chain model, competitive advantage and competitive strategy, different types of ecommerce like B2B, B2C, C2C, C2B,G2C Technology in ecommerce: An overview of the internet, basic network architecture and the layered model, internet architecture, network hardware and software considerations, intranets and extranets ,The making of world wide web, web system architecture, ISP, URL's and HTTP, cookies.	M1	5

CO2	Building and hosting your website: choosing an ISP, registering a domain name, web promotion, internet marketing techniques, e-cycle of internet marketing, personalization, mobile agents, tracking customers, customer service, CRM and e-value Web page design using HTML and CSS: Overview of HTML, basic structure of an HTML document, basic text formatting, links, images, tables, frames, form and introduction to CSS.	M2	8
CO3	Security threats: Security in cyberspace, kinds of threats and crimes: client threat, communication channel threat, server threat, other programming threats, frauds and scams Basic cryptography for enabling security in ecommerce: encryption: public and private key encryption, authentication and trust using digital signature and digital certificates, internet security using VPN, firewalls, SSL Internet payment systems: Features of payment methods, 4C payment methods, electronic money, ACID and ICES test, payment gateway, SET protocol for credit card payment, electronic payment media: ecash and e-wallet, e-check, credit card, debit card, smart card, EFT and ACH, Cyber security.	M3	7
CO4	Business to Business e-commerce: Meaning, benefits and opportunities in B2B, B2B building blocks and their relationship to supply chain management, key B2B models and their main functions, EDI as a B2B tool. Consumer oriented e-commerce: traditional retailing and e-retailing, benefits and key success factors for e-retailing, models for e-retailing like specialized and generalized e-stores, e-mall, direct selling by manufacturer, supplementary distribution channel, e-broker and e-services like web-enabling services, matchmaking services, information selling on the web, entertainment services and auction services.	M4	8
CO5	E-core values: ethical issues, legal issues, taxation issues and international issues.	M5	2

Learning Outcome/ Skills:

1. **Understanding of E-Commerce Concepts and Principles**:

Define and explain the concept of e-commerce and its significance in the digital economy.

2. **Website Development and Design Skills**:

Develop practical skills in designing and developing e-commerce websites using various platforms and tools.

3. **Digital Marketing and Promotion**:

- Gain knowledge of digital marketing strategies and techniques tailored for e-commerce, such as search engine optimization (SEO), social media marketing, email marketing, and content marketing.

4. **E-Commerce Payment Systems and Security**:

Learn about security measures and protocols to protect e-commerce transactions and customer data from cyber threats and fraud.

5. **Inventory Management and Supply Chain Operations**:

Understand strategies for optimizing inventory levels, managing order fulfillment, and improving shipping and delivery logistics.

6. **Customer Relationship Management (CRM)**:

- Learn how to effectively manage customer relationships and provide excellent customer service in the e-commerce context.

7. **Data Analytics and Performance Measurement**:

Gain skills in interpreting data insights and making data-driven decisions to optimize e-commerce operations and marketing strategies.

8. **Legal and Ethical Considerations**:

Develop ethical awareness and adhere to best practices in e-commerce business operations, data management, and customer privacy.

9. **Adaptability and Innovation**:

Develop adaptability and innovation skills to respond to changing market dynamics and customer preferences in the rapidly evolving e-commerce landscape.

10. **Entrepreneurship and Business Acumen**:

Develop strategic thinking, problem-solving, and decision-making skills to identify opportunities, mitigate risks, and drive business growth in the e-commerce sector.

Module Number	Content	Total Hours	% of questions	Bloom Level (applicable)	Remarks, if any
M1	Introduction to ecommerce	5	20%	1,2	NA
M2	Building and hosting your website	8	30%	1,2,3	NA
M3	Security threats	7	20%	1,2,3	NA
M4	Business to Business e-commerce	8	10%	1,2,3	NA
M5	E-core values	2	20%	2,3	NA
Total Theory		30	100		
Tutorial		8			
Total		38			